Ashutosh Adhikari

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Speech ML engineer. Currently working on a single 120M-parameter NEST FastConformer model that covers 22 languages with automatic language identification. Has co-trained publicly released ASR models and written inference servers for client deployments.

Education

The University of Texas at Austin Aug 2026 – Present
M.S. in Computer Science

NMIMS MPSTME Aug 2021 – May 2025
B.Tech (Hons.), Artificial Intelligence *GPA: 3.77/4.0 · Dean's List (top 10%)*

Experience

Generative AI Engineer, BharatGen (IIT Bombay) May 2025 – Present
Promoted from AI Software Engineer, Nov 2025

- Currently working on a single NEST FastConformer model (120M parameters) covering **22 languages** with automatic language identification (unreleased); now adding further capabilities to the same model.
- Contributed to Shrutam-2, a publicly released LLM-based ASR system for 12 Indic languages (Conformer encoder, frozen LLM decoder, SMEAR-MoE projector).
- Co-trained Shrutam-HindiASR-1.0, a released E-Branchformer model reaching **11.58 WER** on Vistaar Hindi (vs. 13.6 for IndicWhisper); wrote the `lazy-infer` ESPnet branch used as the model's official inference dependency.
- Wrote inference servers for clients, using Silero VAD and batched inference.
- Curated and quality-filtered **40,000+ hours** of multilingual speech using WER-based scoring with IndicConformer; ran multi-GPU, multi-node training.
- Built WER/CER evaluation pipelines across Whisper, Conformer, and Branchformer models.

Machine Learning Intern, IIT Bombay Jan 2025 – May 2025
Speech recognition & synthesis research

- Explored Conformer architectures for ASR in ESPnet, and tracked and benchmarked the experiments.
- Worked on VITS-based text-to-speech alongside the ASR work.
- Built a FastAPI annotation tool to collect QMOS/SMOS listening-test ratings for speech model evaluation.
- Built and hosted live demos for the whole team's ASR and TTS models.

Technical Skills

Speech & ASR: NVIDIA NeMo, ESPnet, Conformer, E-Branchformer, LLM-based ASR, CTC, RNN-T, Whisper, VITS, Silero VAD, torchaudio
Modeling & Training: PyTorch, Hugging Face Transformers, SMEAR-MoE, multi-GPU and multi-node training, CUDA, quantization/QAT
Infrastructure: SLURM, Docker, Linux, AWS (EC2/S3), FastAPI, Weights & Biases, Git
Languages: Python, Bash

Under Review

I. Palicha, **A. V. Adhikari**, R. Kumar, S. Santhiappan, G. Ramakrishnan. "SpanMoE-ASR: Span-Constrained Mixture-of-Experts for Controllable Indic ASR Post-Correction." Under review, EACL 2027 Industry Track.

N. Mandloi, M. Kopikar, A. Singh, **A. Adhikari**, K. C. Narendra. "A Learnable Router Layer for Low-Rank Mixture-of-Experts in Hindi-English Code-Switching ASR." Under review, ICASSP 2027.